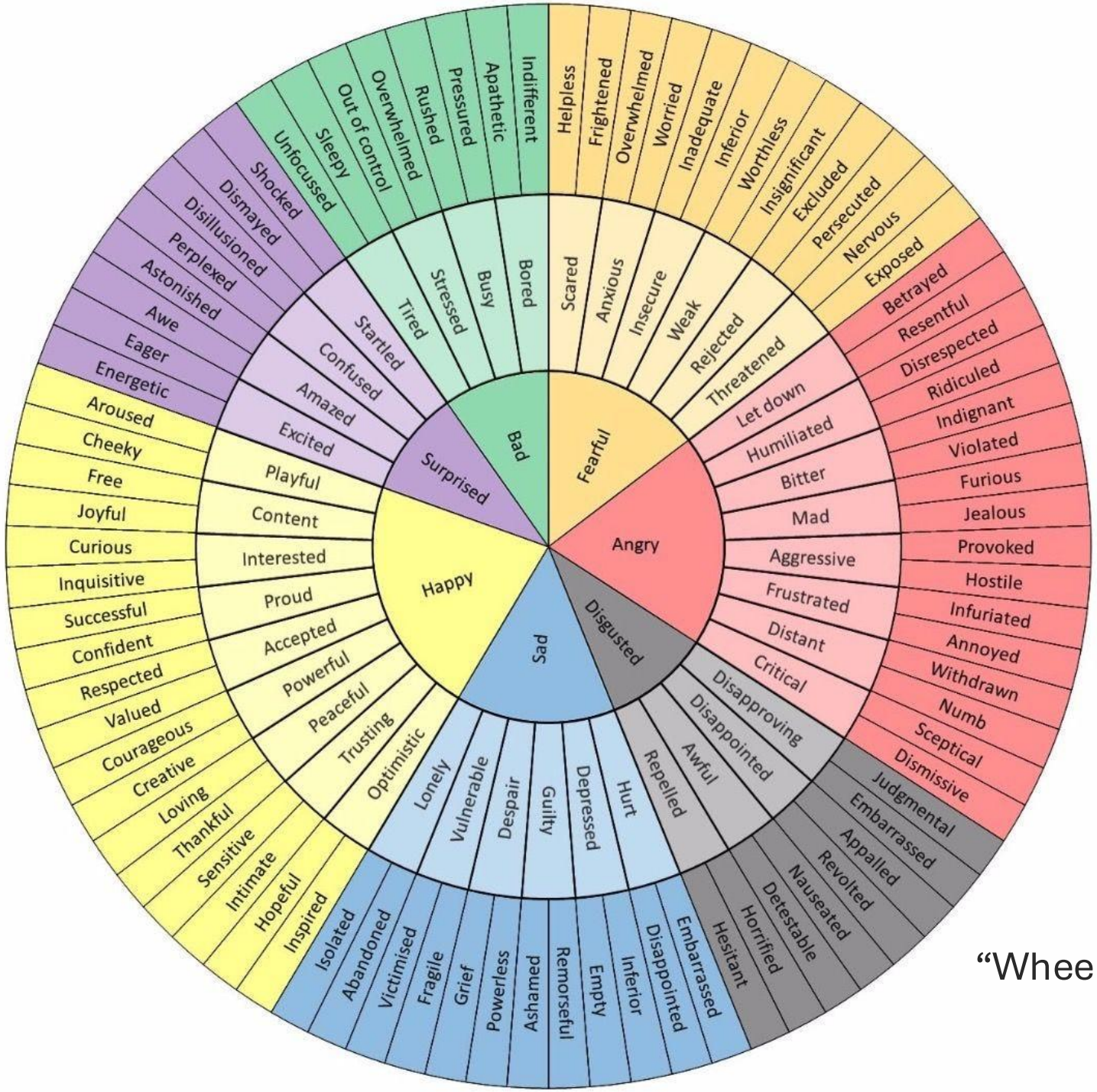
User Tasks – Analysis and Sensemaking

Arpit Narechania

Agenda

- Recap: High-level purpose of visualizations
- Analysis Task Taxonomies
- Sensemaking Loop
- Sneak peek into Presentation and Communication

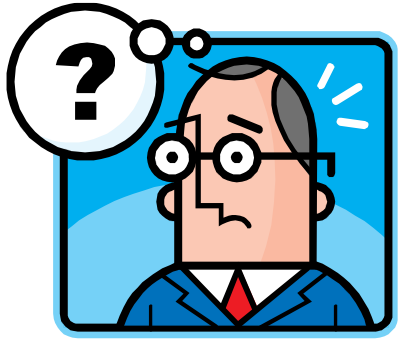
How is everyone feeling today?



"Wheel of Emotions"

What for?

- In order to build better visualizations, we need to understand what people might use them for?
- What do they want to accomplish?



Challenge

Coffee shop sales

Identify potential goals, tasks, and analytic questions that someone holding this data set may have

region	state	category	type	caffeination	sales	profit
Central	CO	Coffee	Amaretto	Regular	219	94
Central	CO	Coffee	Columbian	Regular	190	68
Central	CO	Coffee	Decaf Irish Cream	Decaf	234	101
Central	CO	Tea	Green Tea	Regular	100	30
Central	CO	Espresso	Caffe Mocha	Regular	134	54
Central	CO	Espresso	Decaf Espresso	Decaf	180	53
Central	CO	Herbal Tea	Chamomile	Decaf	341	99
Central	CO	Herbal Tea	Lemon	Decaf	150	0
Central	CO	Herbal Tea	Mint	Decaf	140	33
Central	CO	Tea	Darjeeling	Regular	130	17
Central	CO	Tea	Earl Grey	Regular	140	36
East	CT	Tea	Darjeeling	Regular	123	47
East	CT	Tea	Green Tea	Regular	94	35
East	CT	Espresso	Decaf Espresso	Decaf	130	30
East	CT	Espresso	Caffe Mocha	Regular	180	7
East	CT	Coffee	Columbian	Regular	310	115
East	CT	Herbal Tea	Lemon	Decaf	190	40
East	CT	Herbal Tea	Mint	Decaf	170	47
East	FL	Tea	Darjeeling	Regular	80	28
East	FL	Tea	Green Tea	Regular	81	28
East	FL	Espresso	Decaf Espresso	Decaf	180	50
East	FL	Espresso	Caffe Mocha	Regular	190	9
East	FL	Coffee	Decaf Irish Cream	Decaf	200	67
East	FL	Coffee	Columbian	Regular	210	68
East	FL	Herbal Tea	Chamomile	Decaf	110	34
East	FL	Herbal Tea	Lemon	Decaf	120	13
East	FL	Herbal Tea	Mint	Decaf	150	39

coffee

Tasks

region	state	category	type	caffeination	sales	profit
Central	CO	Coffee	Amaretto	Regular	219	94
Central	CO	Coffee	Columbian	Regular	190	68
Central	CO	Coffee	Decaf Irish Cream	Decaf	234	101
Central	CO	Tea	Green Tea	Regular	100	30
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East	CT	Herbal Tea	Mint	Decaf	170	47
East	FL	Tea	Darjeeling	Regular	80	28
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East	FL	Espresso	Decaf Espresso	Decaf	180	50
East	FL	Espresso	Caffe Mocha	Regular	190	9
East	FL	Coffee	Decaf Irish Cream	Decaf	200	67
East	FL	Coffee	Columbian	Regular	210	68
East	FL	Herbal Tea	Chamomile	Decaf	110	34
East	FL	Herbal Tea	Lemon	Decaf	120	13
East	FL	Herbal Tea	Mint	Decaf	150	39

coffee

Tasks

- Where are most sales coming from (geo)?
- Which type is most profitable?
- Is there a correlation between caffeine & profit?
- Correlation between location & caffeine?
- Compare tea vs coffee sales.
- Which area has most profit?
- Which category has largest margins?
- Any diffs in sales vs. profit?

Follow-on

- What are the (types of) tasks being done here?
  - Abstract away the domain

High-level Purpose

- Main (high-level) visualization goals/objectives

Data analysis	Exploratory
Data presentation	Explanatory

1. Analysis

Explore the data
Assess a situation
Determine how to proceed
Decide what to do

Frequent analysis goals

Show many/all variables

Illustrate overview and detail

Facilitate comparison

Display may not be easy to interpret at first

2. Presentation

Communicate data and ideas

Explain and inform

Provide evidence and support

Influence and persuade

Frequent presentation goals

Clarify
Focus
Highlight
Simplify
Persuade

May just show a few variables and/or a subset of the data cases

More Complex Than That?

Exploratory analysis - Deriving hypotheses from an unknown dataset. It is often equated with an undirected search.

Confirmatory analysis - Testing found or assumed hypotheses about a dataset. It is sometimes described as a directed search.

Presentation - Describing and exhibiting confirmed analysis results.

Schulz et al
TVCG (InfoVis) '13

Research Goal

Can we characterize the set of tasks that people want to accomplish with visualizations?

Perhaps create a taxonomy of tasks?

Early Taxonomy

- Wehrend & Lewis created a low-level, domain independent taxonomy of user tasks in visualization environments

- Eleven basic actions
  - identify, locate, distinguish, categorize, cluster, distribution, rank, compare within relations, compare between relations, associate, correlate

Wehrend & Lewis
Vis '90

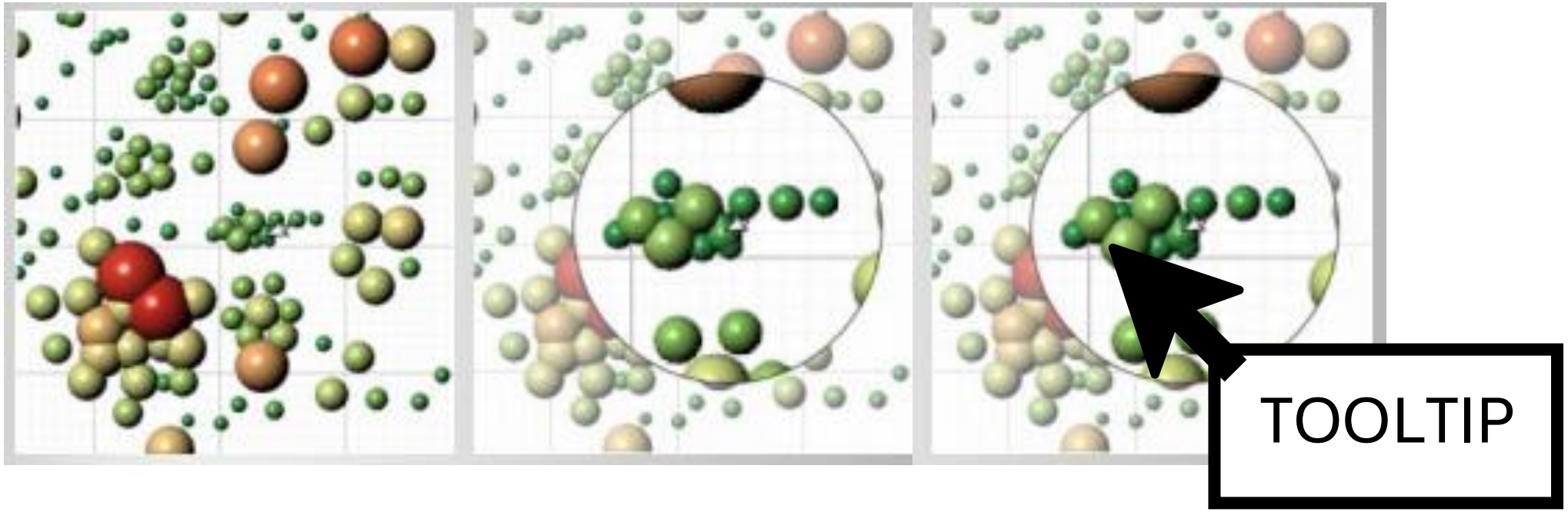
InfoVis Mantra

- Shneiderman proposed "task × data type" taxonomy to understand what people do with visualization
- Mantra:

  “Overview first, zoom and filter, then details on demand”

- Design paradigm for infovis systems

Shneiderman
VL ‘96


TOOLTIP

Can we be more precise?

Focus on lower-level tasks
Be empirically driven

Low-Level Components of Analytic Activity in Information Visualization

Robert Amar, James Eagan, and John Stasko

College of Computing/GVU Center
Georgia Institute of Technology

ABSTRACT

Existing system-level taxonomies of visualization tasks are geared more towards the design of particular representations than the facilitation of user analytic activity. We present a set of ten low-level analysis tasks that largely capture people's activities while employing information visualization tools for understanding data. To help develop these tasks, we collected nearly 200 sample questions from students about how they would analyze five particular data sets from different domains. The questions, while not being totally comprehensive, illustrated the sheer variety of analytic questions typically posed by users when employing information visualization systems. We hope that the presented set of tasks is useful for information visualization system designers as a kind of common substrate to discuss the relative analytic capabilities of the systems. Further, the tasks may provide a form of checklist for system designers.

CR Categories and Subject Descriptors: H.5.0 [Information Interfaces and Presentation]: General; J.0 [Computer Applications]: General

Additional Keywords: Analytic activity, taxonomy, knowledge discovery, design, evaluation.

1 INTRODUCTION

Information visualization research, especially that dealing with the automatic generation of information presentations [10,15], has produced several taxonomies of system tasks that map visualization operations to user cognitive processes. In one sense, these taxonomies might be considered low-level task taxonomies or hierarchies, since they form part of a compositional language upon which automatic generation systems build higher-order externalizations of data.

However, considering these taxonomies as a basis upon which to build models of analytic activity is made difficult by their origins. While their elements can be algorithmically composed into presentations, the composition process itself is ad-hoc, relying on a designer's own insight and expressive capability within a particular tool. These taxonomies reflect this system-

With the aim of generating an actionable means for supporting analytic activity, we wish to rethink some of the lower-level task taxonomies that focus on a generated presentation as an end result. In general, information visualization can benefit from understanding the tasks that users accomplish while doing actual analytic activity. Such understanding achieves two goals: first, it aids designers in creating novel presentations that amplify users' analytic abilities; second, it provides a common vocabulary for evaluating the abilities and affordances of information visualization systems with respect to user tasks.

We argue that a stronger focus on user tasks and analytic activities in information visualization is necessary as current tools do not seem to support analytic activity consistently. A 2004 study by Saraiya and North found that insights generated from tools used to visualize gene expression data were not generally valuable according to domain experts [11]. Systems such as IN-SPIRE [7] support analytic activities within the domain of document search but may not generalize across domains. Current tools may not even support representational activity very well; consider, for example, the Kobsa study showing only 68-75% accuracy on relatively simple tasks during commercial tool evaluation [8].

1.2 The Nature of Analytic Activity

User analysis questions and tasks as part of analytic activity typically range from broader, "high-level" goals to much more specific, "low-level" inquiries. For example, a person studying the history of motion picture films may have "high-level", uncertainty-tinged knowledge goals such as understanding trends in popularity over time or determining how to predict which movies will win Academy Awards. In the process of acquiring this knowledge, the person may generate more specific, low-level queries such as identifying the Academy Award-winning pictures of the past ten years and determining whether or not movie length correlates to the film's popularity.

It is this latter set of questions, more specific and focused in nature, on which we focus on in this article. In particular, we are interested in generating a relatively small set of question types that encompasses the set of user inquiries made while working

Amar, Eagan, & Stasko
InfoVis '05

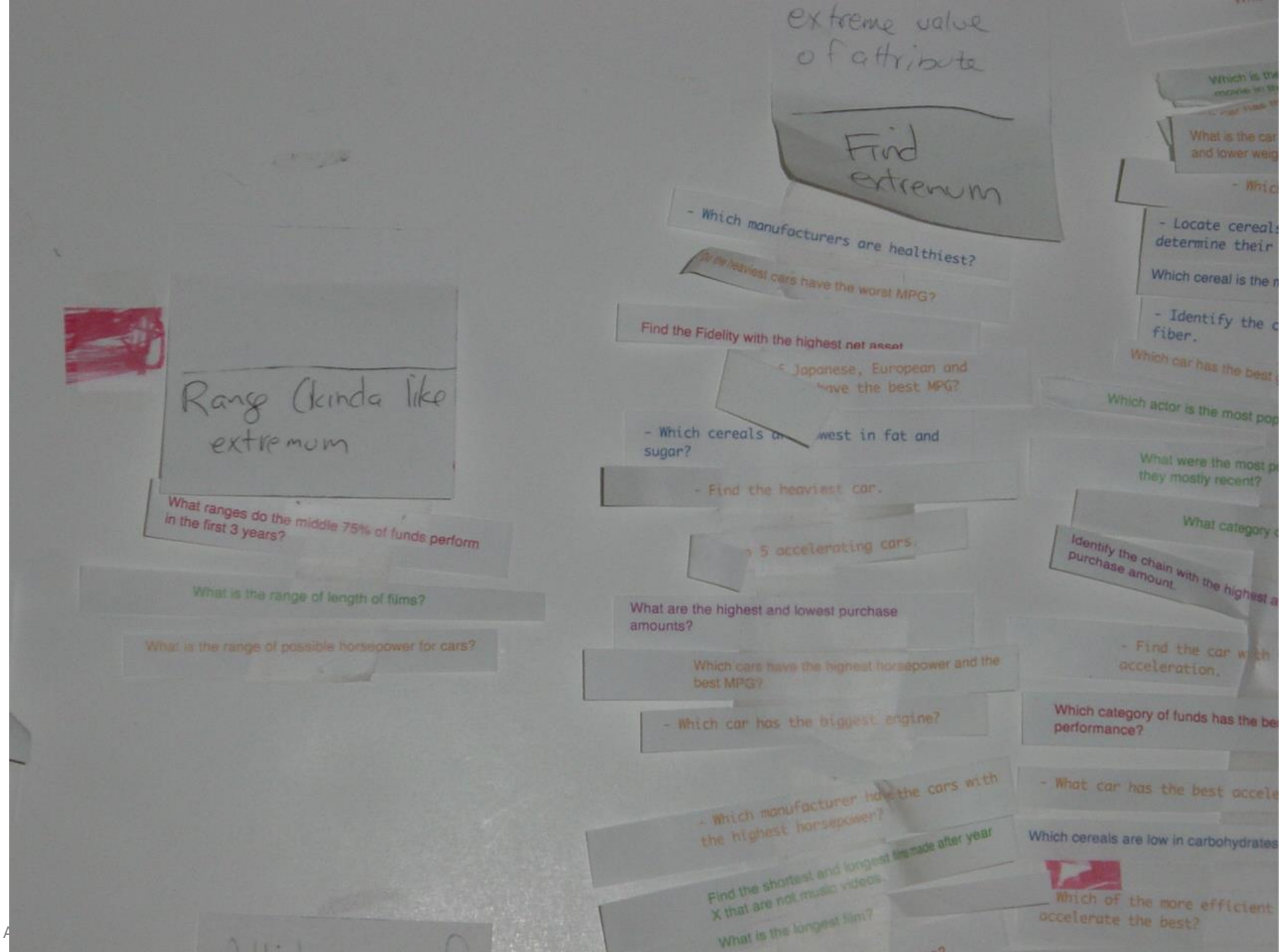
Background

- Use “commercial tools” class assignment from a class at Georgia Tech in early 2000s
- Students generate questions to be answered using commercial infovis systems
- Data sets:

Domain	Data cases	Attributes	Questions Generated
Cereals	78	15	107
Mutual funds	987	14	41
Cars	407	10	153
Films	1742	10	169
Grocery surveys	5164	8	126

- Generated 596 total analysis questions/tasks

Find Attr.
Distribution
Range charts the extremum


extreme value of attribute
Find extremum
Range (kinda like extremum
- Which manufacturers are healthiest?
Find the Fidelity with the highest net asset
What ranges do the middle 75% of funds perform in the first 3 years?
What is the range of length of films?
What is the range of possible horsepower for cars?
- Find the heaviest car.
What are the highest and lowest purchase amounts?
Which cars have the highest horsepower and the best MPG?
- Which car has the biggest engine?
Find the shortest and longest films made after year X that are not music videos.
What is the longest film?
Which actor is the most pop
What were the most p they mostly recent?
What category
Identify the chain with the highest a purchase amount.
Which category of funds has the be performance?
Which cereals are low in carbohydrates
Which of the more efficient accelerate the best?

SAVE
Please do not erase
Organizational operations
(filter, cluster, categ.)
vs.
transformation ops.
(avg., count, ...)
1-1 n-1
find extremum
extreme mum
aggregate
Filter
Cluster
Categorize
What is the car w/ highest MPG?
What director has won the most awards?
What film
What Robin Williams film has release date ↑?
→ Case(s) at extreme of range of attr(s)
look-up value(s)
(find entity)
Simple Comparison
→ compare scalar values
→ compare the avg MPG of American and Japanese cars.
→ What is mpg of Audi TT?
→ How long is Gone w/ Wind?
→ Given case(s), what are some attributes?
Computed meta-value
→ Computed meta-value
→ What is the average calorie content of Post cereals?
→ What is the gross income of all stores combined?
Filter
→ What Kellogg's cereals have > 3g fiber?
→ What comedies have won awards?
→ mapping from attr → case(s)
→ What funds underperformed S&P 500?
Outliers
→ "Iconic" values
→ Are there any outliers to the typical relationship?
→ Are there any outliers in prices?
Extremum of X
→ extremum of a set of cardinals
→ What mfg has largest # of cereals?
→ Who starred in most films in 1988?
count
How many films are 127 mins long?
How many manufacturers of cars are there?
→ Find cardinality of set of cases
Categorical Correlation
→ correlation w/ categorical variables
→ Is there a correlation between Country of origin and MPG?
→ Do different genders have a preferred payment method?
Correlation
→ correlation of 2 numeric attributes
→ Is there a correlation between CPRS and fat?
→ Are [illegible] films more popular?
Range
→ range of an attribute in a set of cases
→ What is the range of the length of films?
→ What is the range of horsepower of cars?
Distribution
→ Characterization of distribution of an attribute over set of cases
→ What is the distribution of calbs in cereals?
→ What is age distribution of shoppers?
Sort
→ order a set of cases by some metric
→ Order the cars by weight
→ Rank the cereals by calories
Browse
→ loosely guided exploration of data.
→ Do any variables correlate w/ fat?
→ Are there trends among countries of origin?
Clustering
→ Find clusters of attribute values in a set of cases
→ Are there groups of cereals w/ similar fat, cal, sugar?
→ Is there a cluster of typical film lengths
trend vs. correlate

Terminology

- *Data case* – An entity in the data set
- *Attribute* – A value measured for all data cases
- *Aggregation function* – A function that creates a numeric representation for a set of data cases (eg, average, count, sum)

1. Retrieve Value

General Description:
Given a set of specific cases, find attributes of those cases.

Examples:
- What is the mileage per gallon of the Audi TT?
- How long is the movie Gone with the Wind?

2. Filter

General Description:
Given some concrete conditions on attribute values, find data cases satisfying those conditions.

Examples:
- What Kellogg's cereals have high fiber?
- What comedies have won awards?
- Which funds underperformed the SP-500?

3. Compute Derived Value

General Description:

Given a set of data cases, compute an aggregate numeric representation of those data cases.

Examples:

- What is the gross income of all stores combined?
- How many manufacturers of cars are there?
- What is the average calorie content of Post cereals?

4. Find Extremum

General Description:

Find data cases possessing an extreme value of an attribute over its range within the data set.

Examples:

- What is the car with the highest MPG?
- What director/film has won the most awards?
- What Robin Williams film has the most recent release date?

5. Sort

General Description:
Given a set of data cases, rank them according to some ordinal metric.

Examples:
- Order the cars by weight.
- Rank the cereals by calories.

6. Determine Range

General Description:
Given a set of data cases and an attribute of interest, find the span of values within the set.

Examples:
- What is the range of film lengths?
- What is the range of car horsepowers?
- What actresses are in the data set?

7. Characterize Distribution

General Description:

Given a set of data cases and a quantitative attribute of interest, characterize the distribution of that attribute's values over the set.

Examples:

- What is the distribution of carbohydrates in cereals?
- What is the age distribution of shoppers?

8. Find Anomalies

General Description:

Identify any anomalies within a given set of data cases with respect to a given relationship or expectation, e.g. statistical outliers.

Examples:

- Are there any outliers in protein?
- Are there exceptions to the relationship between horsepower and acceleration?

9. Cluster

General Description:
Given a set of data cases, find clusters of similar attribute values.

Examples:
- Are there groups of cereals w/ similar fat/calories/sugar?
- Is there a cluster of typical film lengths?

10. Correlate

General Description:
Given a set of data cases and two attributes, determine useful relationships between the values of those attributes.

Examples:
- Is there a correlation between carbohydrates and fat?
- Is there a correlation between country of origin and MPG?
- Do different genders have a preferred payment method?
- Is there a trend of increasing film length over the years?

Compound tasks

- "Sort the cereal manufacturers by average fat content"
  Compute derived value; Sort

- "Which actors have co-starred with Julia Roberts?"
  Filter; Retrieve value

Reflection: What questions were left out?

- Basic math
  “Which cereal has more sugar, Cheerios or Special K?”
  “Compare the average MPG of American and Japanese cars.”

- Uncertain criteria
  “Does cereal (X, Y, Z…) sound tasty?”
  “What are the characteristics of the most valued customers?”

- Higher-level tasks
  "How do mutual funds get rated?"
  “Are there car aspects that Toyota has concentrated on?”

- More qualitative comparison
  “How does the Toyota RAV4 compare to the Honda CRV?”
  “What other cereals are most similar to Trix?”

Concerns/Limitations

- InfoVis tools may have influenced students' questions
- Graduate students as group being studied
  - How about professional analysts?
- Subjective – Not an exact science
- Data was really quantitative so may get a different set of tasks for relational/graph data
  - See Lee et al, BELIV '06

Contributions

- Set of grounded low-level analysis tasks
- Potential use of tasks as a language/vocabulary for comparing and evaluating infovis systems

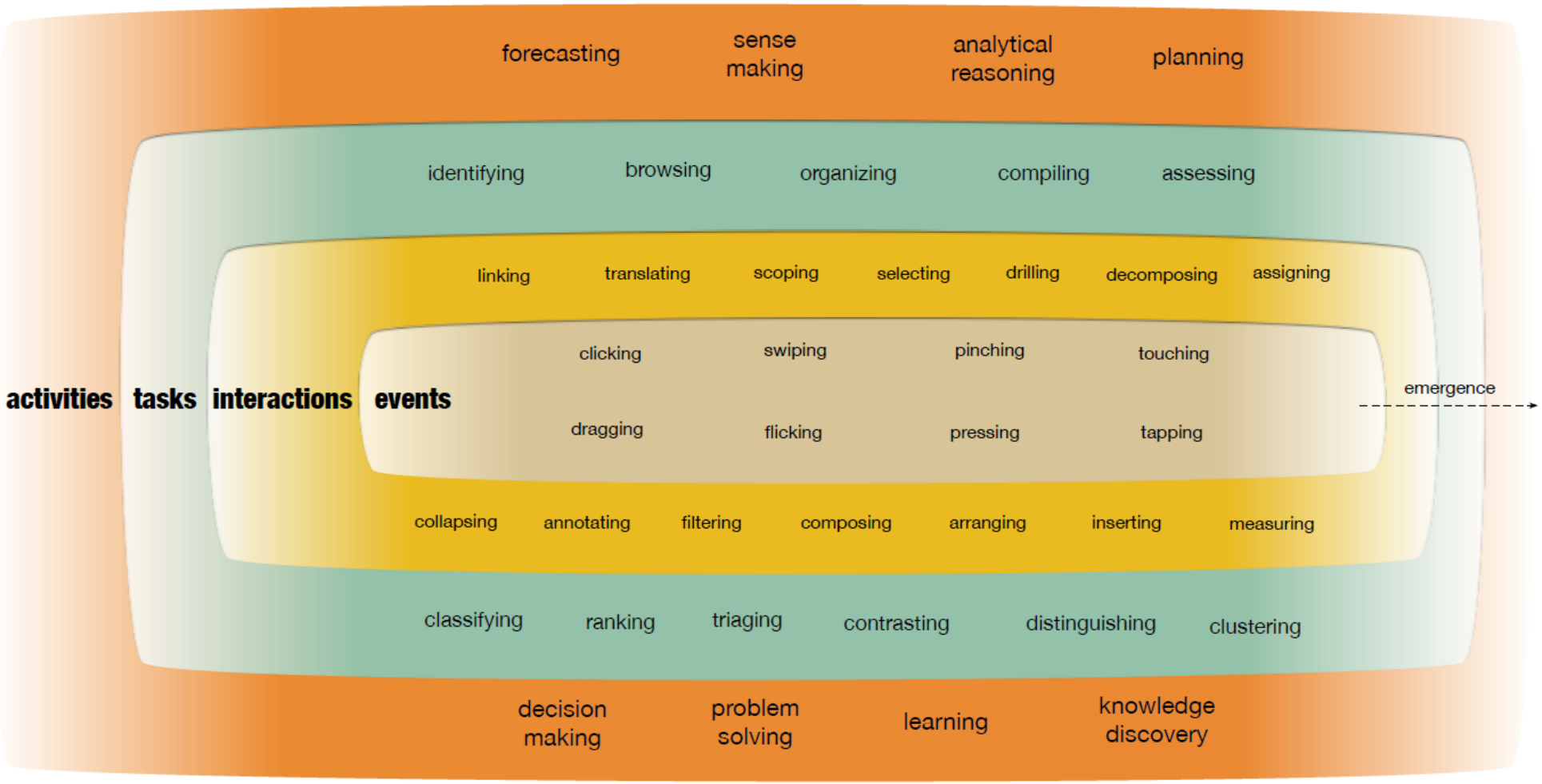
All About the Level

- High – Analysis, exploration
- Low – Correlation, Outlier finding

A nice conceptual model…

Visual Analytic Activity

Sedig, Parsons, Babanski
JMPT '12



Abstract Tasks

Framework/Typology of abstract visualization tasks

A Multi-Level Typology of Abstract Visualization Tasks

Matthew Brehmer and Tamara Munzner, *Member, IEEE*

Abstract—The considerable previous work characterizing visualization usage has focused on low-level tasks or interactions and high-level tasks, leaving a gap between them that is not addressed. This gap leads to a lack of distinction between the ends and means of a task, limiting the potential for rigorous analysis. We contribute a multi-level typology of visualization tasks to address this gap, distinguishing *why* and *how* a visualization task is performed, as well as *what* the task inputs and outputs are. Our typology allows complex tasks to be expressed as sequences of interdependent simpler tasks, resulting in concise and flexible descriptions for tasks of varying complexity and scope. It provides abstract rather than domain-specific descriptions of tasks, so that useful comparisons can be made between visualization systems targeted at different application domains. This descriptive power supports a level of analysis required for the generation of new designs, by guiding the translation of domain-specific problems into abstract tasks, and for the qualitative evaluation of visualization usage. We demonstrate the benefits of our approach in a detailed case study, comparing task descriptions from our typology to those derived from related work. We also discuss the similarities and differences between our typology and over two dozen extant classification systems and theoretical frameworks from the literatures of visualization, human-computer interaction, information retrieval, communications, and cartography.

Index Terms—Typology, visualization models, task and requirements analysis, qualitative evaluation

1 INTRODUCTION

Consider a person who encounters a choropleth map while reading a blog post in the aftermath of last year's American presidential election. This particular map is static and visually encodes two attributes, candidate and margin of victory, encoded for each state using a bivariate colour mapping. This person decides to compare the election results of Texas to those of California, motivated not by an explicit need to generate or verify some hypothesis, nor by a need to present the visualization to an audience, but rather by a casual interest in American politics and its two most populous states. How might we describe this person's *task* in an abstract rather than domain-specific way?

According to the nested model for visualization design and validation [43], abstract tasks are domain- and interface-agnostic operations performed by users. Disappointingly, there is little agreement as to the appropriate granularity of an abstract task among the many extant classifications of user behaviour in the visualization, human-computer interaction, cartography, and information retrieval literature [2, 3, 5, 10, 11, 12, 13, 14, 15, 19, 23, 29, 31, 37, 39, 42, 50, 51, 56, 57, 59, 61, 64, 66, 72, 73, 75, 78, 82, 83]. One of the more frequently cited of these [2] would classify the above example as being a series of value retrieval tasks. This low-level characterization does not describe the user's context or motivation; nor does take into account prior experience and background knowledge. For instance, a description of this task might differ if the user was unfamiliar with American geography: the user must locate and identify these states before comparing their values. Conversely, high-level descriptions of exploratory data analysis and presentation emanating from the sensemaking literature [3, 11, 31, 51] cannot aptly describe this user's task.

The gap between low-level and high-level classification leaves us unable to abstractly describe user tasks in a useful way [41], even for the simple static visualization in the above example. This gap widens when interactive visualization is considered, and the complexity of its usage is compounded over time. We must move beyond describing a single task in isolation, to a description that designates when one task ends and another begins. To close this gap, visualization tasks must be describable in an abstract way across multiple levels.

The primary contribution of this paper is a multi-level typology of abstract visualization tasks that unites the previously disconnected scopes of low-level and high-level classification systems by proposing multiple levels of linkage between them. Our typology provides a powerful and flexible way to describe complex tasks as a sequence of interdependent simpler ones. While this typology is very much informed by previous work, it is also the result of new thinking and has many points of divergence with previous models. Central to the organization of our typology are three questions that serve to disambiguate the means and ends of a task: *why* the task is performed, *how* the task is performed, and *what* are the task's inputs and outputs. We have found that no prior characterization of tasks satisfactorily answers all of these questions simultaneously at multiple levels of abstraction. Typically, low-level classification systems provide a sense of *how* a task is performed, but not *why*; high-level classification systems are the converse. One major advantage of our typology over prior work is in providing linkage between these two questions. Another advantage is the ability to link sequences of tasks, made possible by the consideration of *what* tasks operate on.

Our typology provides a consistent lexicon for description that supports making precise comparisons of tasks between different visualization tools and across application domains. Succinct and abstract descriptions of tasks are crucial for analysis of visualization usage. This analysis is an essential precursor to the effective design and evaluation of visualization tools, particularly in the context of problem-driven design studies [60]. In these studies, visualization practitioners work with users to identify *why* and *what*, subsequently drawing from their specialized knowledge of visual encoding and interaction techniques to design *how* that task is to be supported [41]. A need for task analysis also arises in visualization evaluation [34], particularly in observational studies of open-ended visualization usage. Our typology provides a multi-level code set for qualitatively describing user behaviour in such studies.

As we expect some readers to be unfamiliar with the context that motivated this work, we begin with a brief discussion of our current inability to succinctly describe and analyze visualization tasks. In Section 3 of this paper, we introduce our multi-level typology of abstract visualization tasks. In Section 4, we demonstrate the benefits of our approach with a detailed case study, in which we describe a sequence of interdependent tasks. In Section 5, we summarize our typology's connections to related work and to its theoretical foundations. In Section 6, we discuss the value and usage of this typology, as well as our plans for its further validation and extension. In Section 7, we offer our concluding remarks.

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Manuscript received 31 March 2013; accepted 1 August 2013; posted online 13 October 2013; mailed on 4 October 2013.
For information on obtaining reprints of this article, please send e-mail to: tvcg@computer.org.

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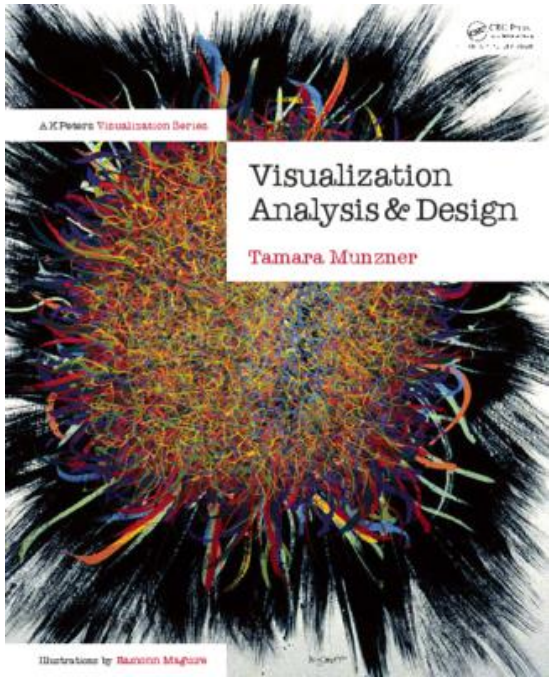
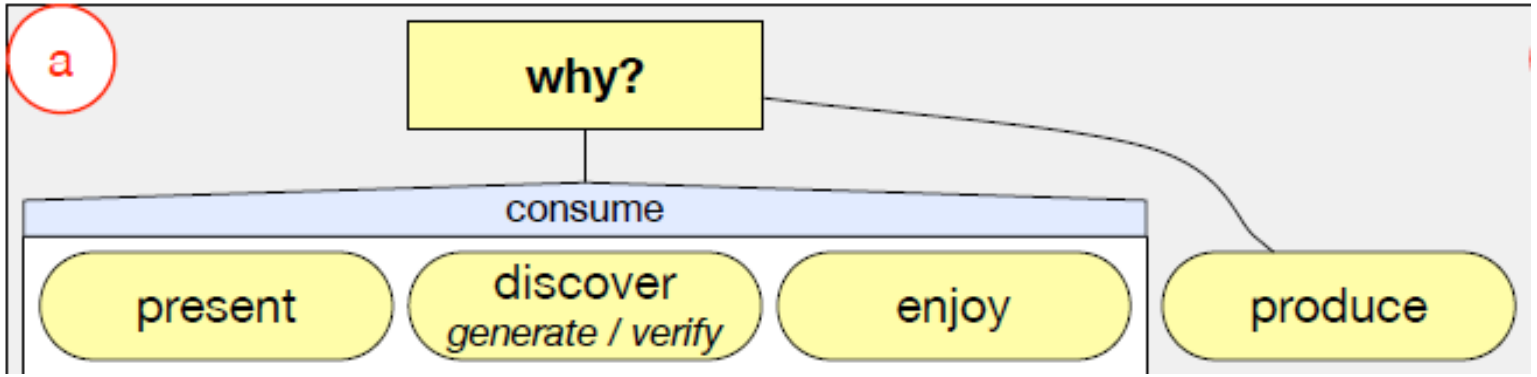
Brehmer & Munzner
TVCG (InfoVis) '13



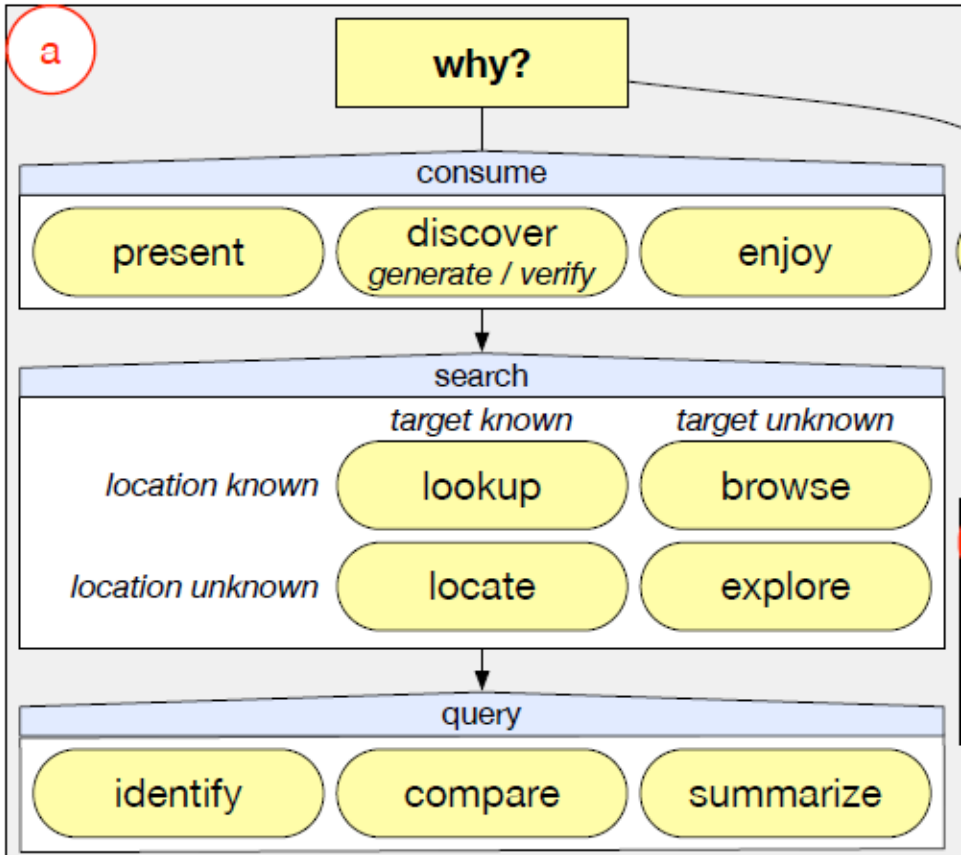
Chapter 3

Why?

What are the top-level categories (answers) to the “Why?” question?



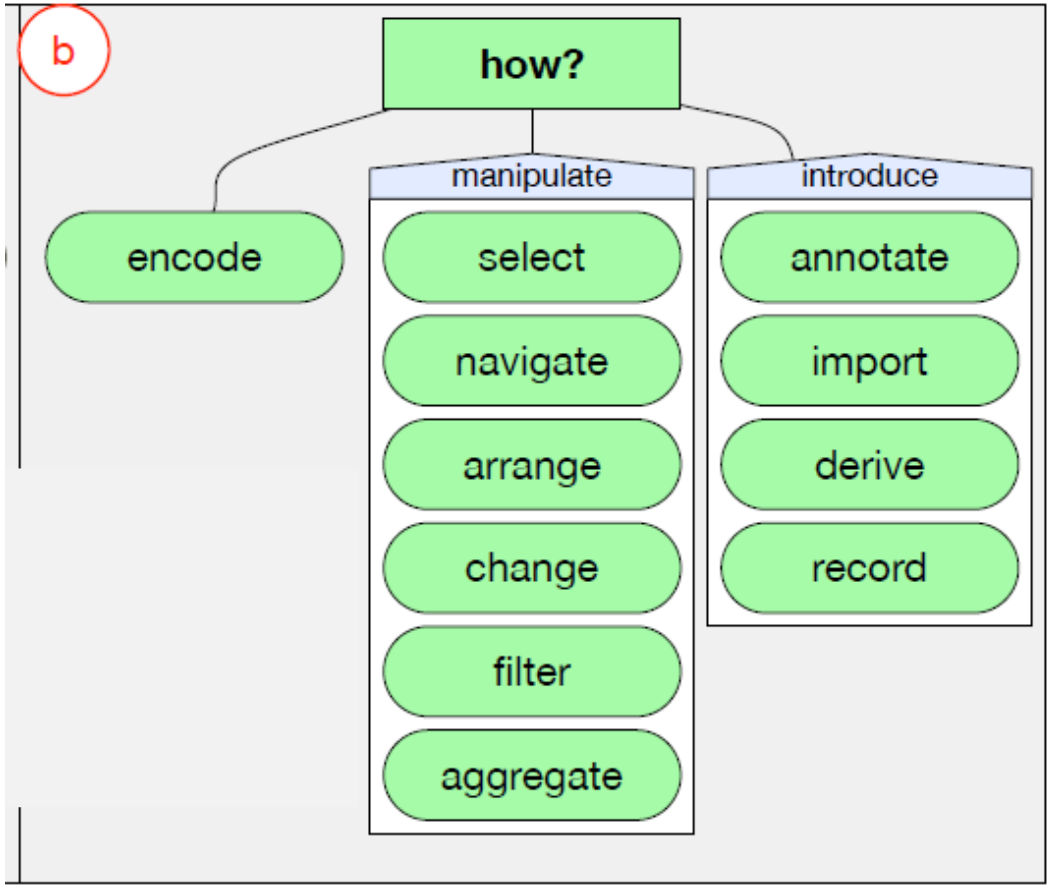
Discover



High to low level

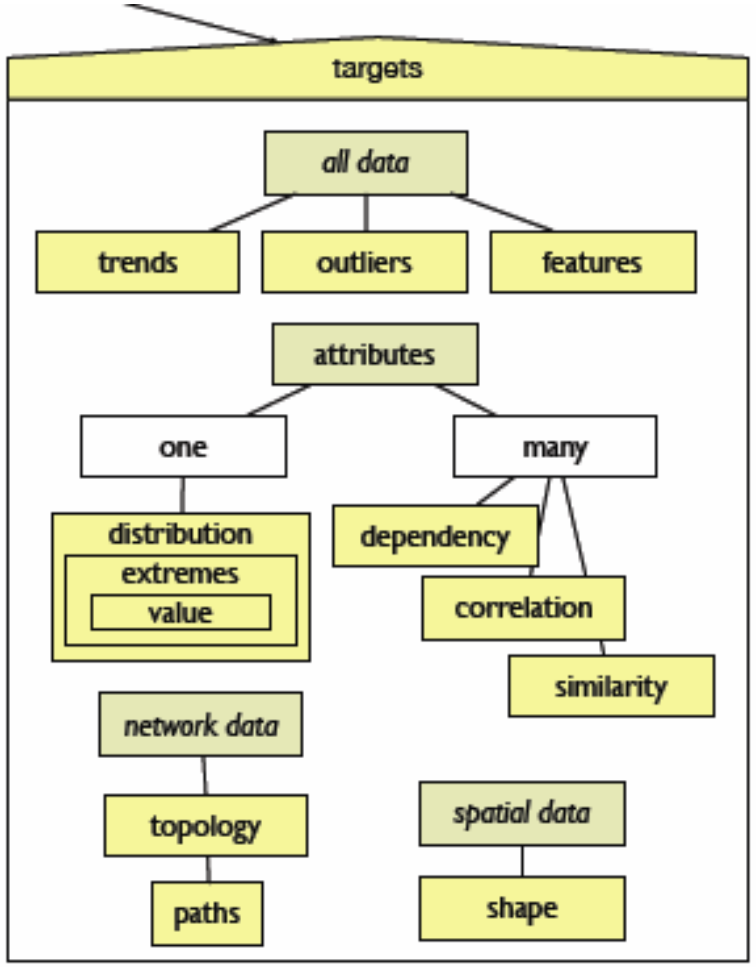
How?

Now the second question



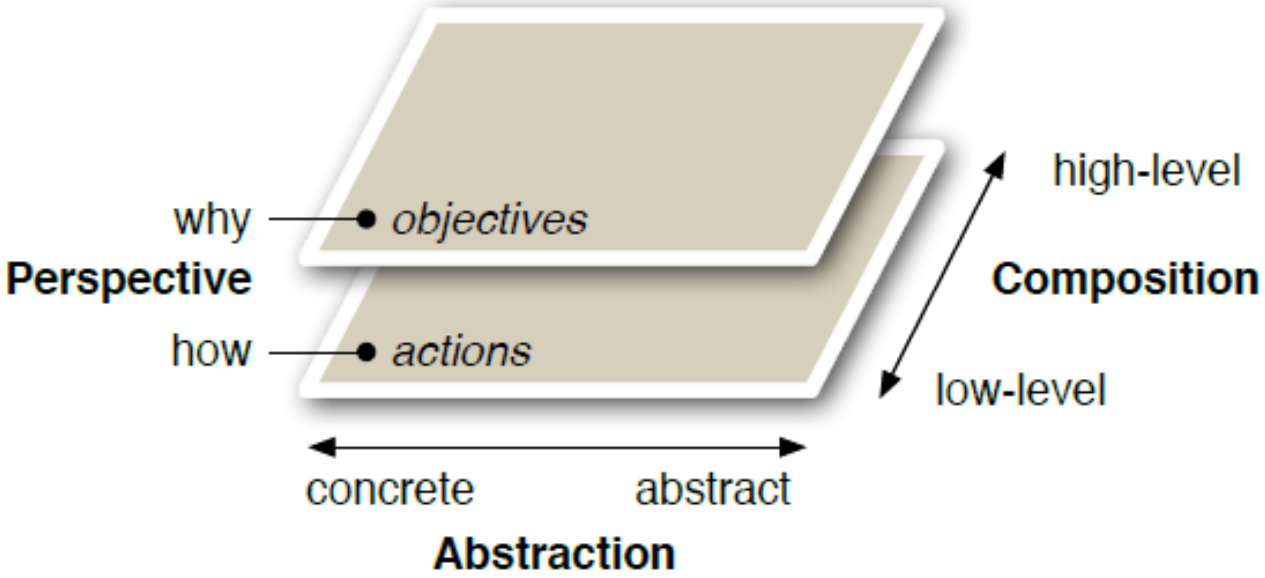
Targets

What are the types of targets?



Task Cube

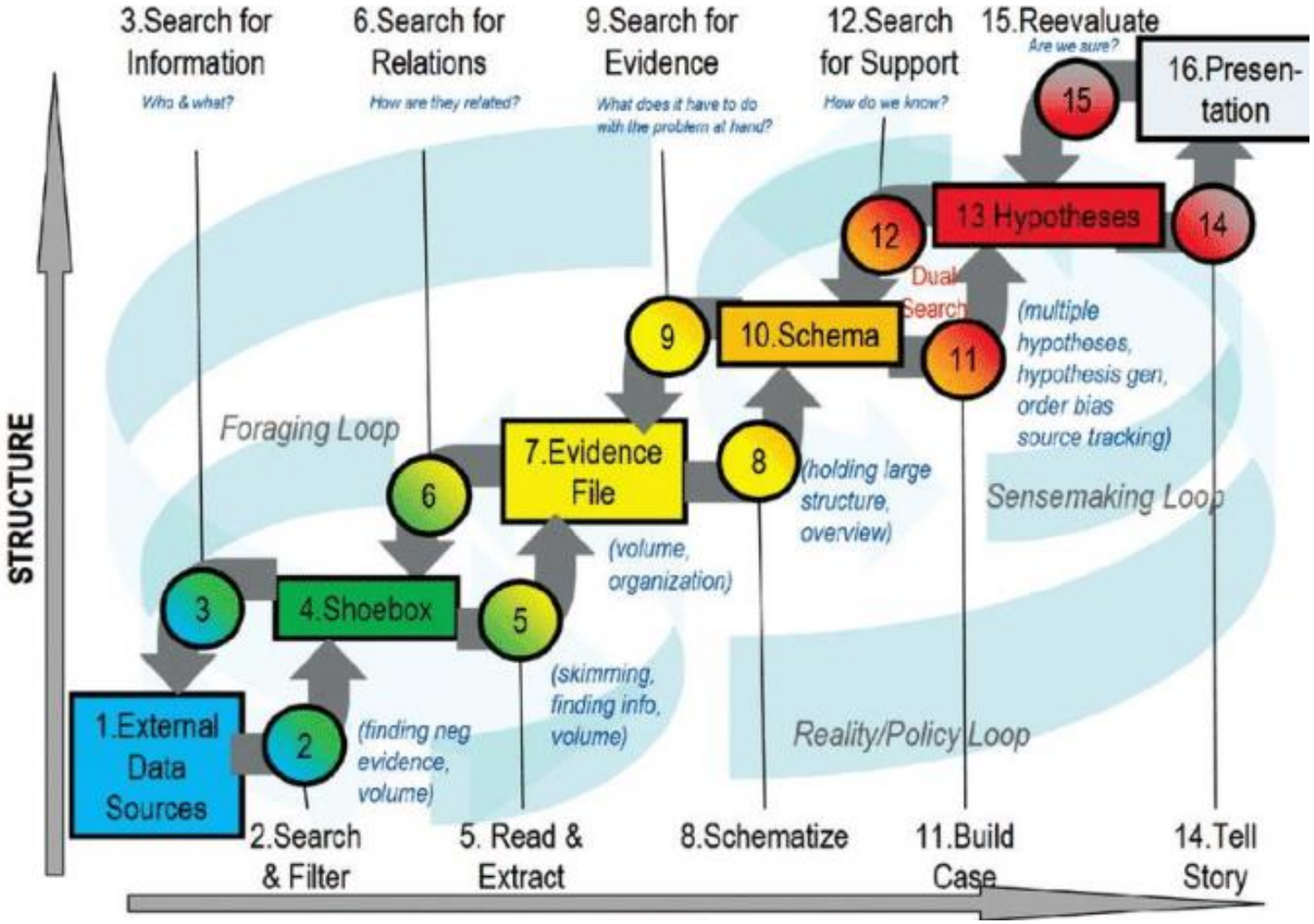
"Task" – Much confusion about term's use, ambiguously used and overloaded in vis community



Rind et al
Information Visualization '15

Sensemaking Model

Pirolli and Card (2005)



Think Carefully

- Preconceptions about visualization's utility
  - Answering specific questions and accomplishing specific analytic tasks
  - Generating unexpected, serendipitous discoveries and insights
    - "Finding a needle in a haystack"

OK, but not what it's best for

Visualization most useful in **exploratory data analysis**

- Don’t know what you’re looking for
- Don’t have a priori questions
- Want to know what questions to ask

So, vis is all about exploration… ?

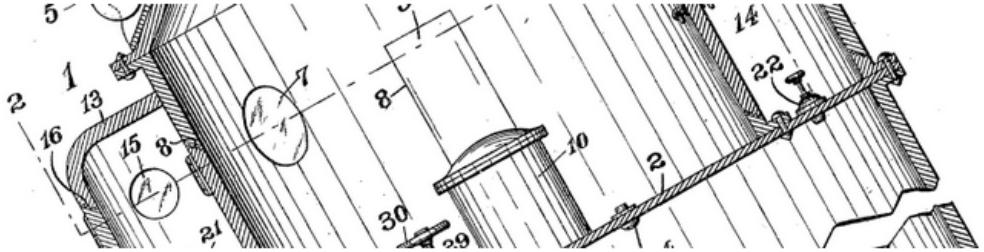
Discuss

https://medium.com/@eytanadar/banning-exploration-in-my-infovis-class-9578676a4705

“No one is paid to explore, they’re paid to find.”

Medium

Eytan Adar Follow
Associate Professor, University of Michigan
Apr 26, 2017 · 9 min read



Patent #1,059,281 (Diving Apparatus for Marine Exploration and the Like)

Banning exploration in my infovis class

I’ve banned the word “explore” from all project proposals in my infovis class. No *explore*. No *exploration*. No *exploratory*. No, you may not create a tool to “allow an analyst to explore the bird strike data.” No, you can’t build a system for “exploration of microarray data.” And, no, you can’t make a framework for “exploratory network analysis.” Just no.

The line that I use on my students is that: *No one is paid to explore, they’re paid to find*. I’m only 10% trying to be clever. Ninety percent, I’m dreading grading the output of projects that feature exploration as an objective.

I got to have lunch with John Tukey many years ago. We talked about birding. I wish we talked about “Exploratory Data Analysis.” For all the clever names he created for things (*software, bit, cepstrum, quefrency*) what’s up with EDA?

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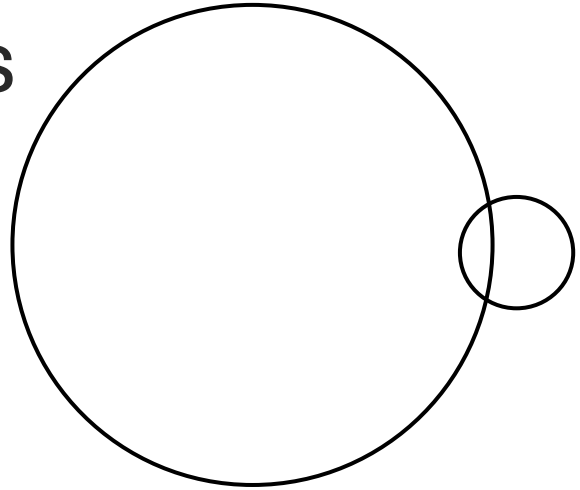
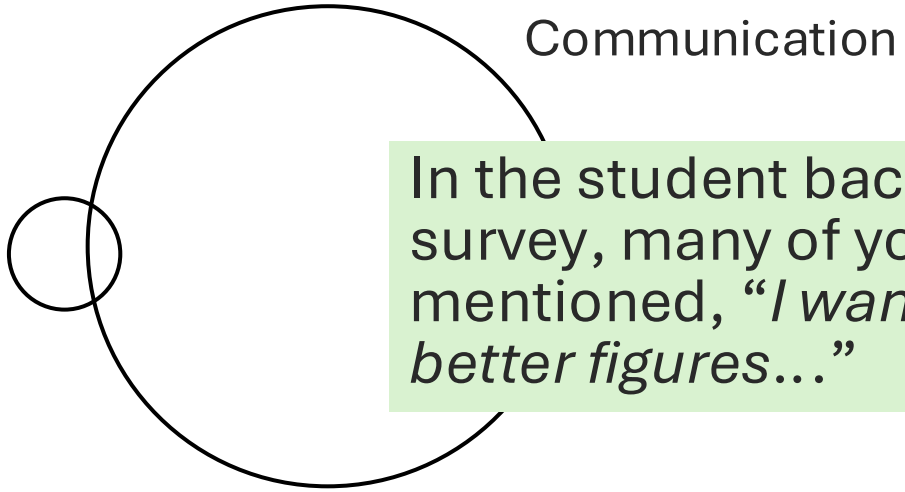
Views of the World

Vis researchers



Everyone else



In the student background survey, many of you have mentioned, "*I want to make better figures…*"

Crucial Point

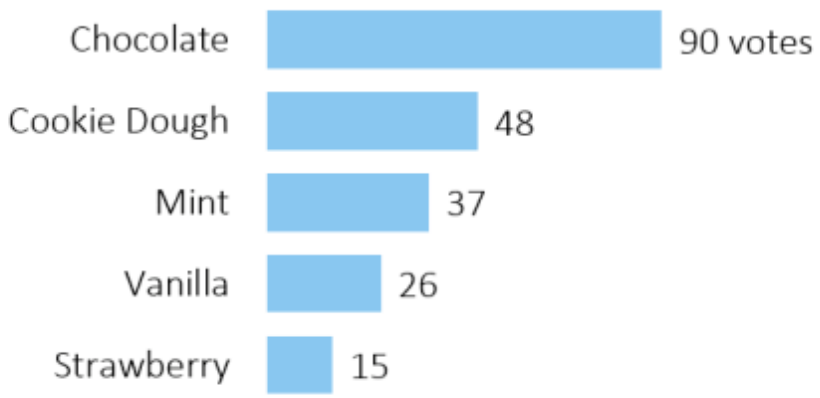
Analytic:
We're searching for the answers and insights from the data
We're investigating

Presentation:
Answers and insights already known
We need to communicate them

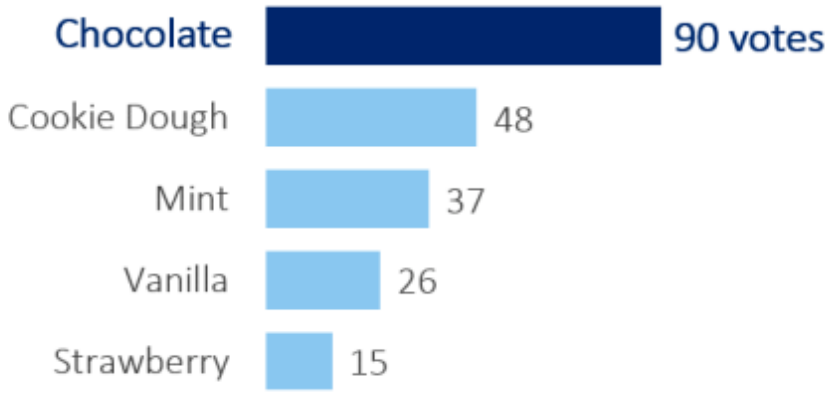
https://depictdatastudio.com/how-to-tell-a-story-with-data-titles-subtitles-annotations-dark-light-contrast-and-selective-labeling/

An Example

Ice cream flavor preferences based on 2014 survey of elementary school students (n=216)



Chocolate was most popular flavor
among elementary students surveyed



Source: 2014 survey of elementary school students (n=216)

Review

- Recap: High-level purpose of visualizations
- Analysis Task Taxonomies
- Sensemaking Loop
- Sneak peek into Presentation and Communication

ToDo

- GP-M2: Update your Team info with preliminary proposal in the Google Sheet.

Next time

- User Tasks: Presentation and Storytelling